JANE DU

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RESEARCH INTERESTS

Machine Learning, Multitask/Multi-Domain Learning, Foundation Models & Large Language Models (LLMs), Reinforcement Learning, Computer Vision, Optimization

EDUCATION

Ph.D. in Computer Science from University of Illinois, Urbana-Champaign **Expected: May 2027**
- Advised by Prof. Arindam Banerjee

M.S. in Computer Science from Cornell University **May 2021**
- Advised by Prof. Karthik Sridharan

B.S. in Computer Science from Cornell University, Minor in Mathematics **May 2019**

PEER-REVIEWED CONFERENCE PUBLICATIONS

J. Du, A. Banerjee, "Computationally Efficient Methods for Invariant Feature Selection with Sparsity", In *Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence* (UAI 2025)

RESEARCH AND AWARDS

Personalized Federated Learning: Communication and Parameter-Efficient Personalization **Ongoing**
- Investigated personalized federated approaches on diverse geospatial data sources (TorchGeo imagery from distinct sensors and regions) to enhance learning under domain heterogeneity
- Studied parameter-efficient personalization schemes (adapter-style heads / lightweight modules) inspired by LLM fine-tuning to reduce communication and memory costs in federated settings
- Evaluated federated fine-tuning of pre-trained transformer backbones (ViT) under heterogeneous client distributions, measuring personalization vs. global performance

Private Personalized Federated Reinforcement Learning **Ongoing**
- Investigated federated reinforcement learning across heterogeneous MDPs that share latent structure, enabling client-specific policy and value adaptation under limited communication
- Analyzed privacy–utility and personalization–generalization tradeoffs for federated policy learning, with implications for privacy-preserving fine-tuning of large sequence models across heterogeneous clients

Computationally Efficient Methods for Invariant Feature Selection with Sparsity **July 2025**
- Developed computationally efficient methods for casual inference
- Extracted causal (invariant) features in overparameterized settings with deep learning models, addressing failures of standard IRM that mistakenly rely on spurious correlations

Nonconvex Constrained Optimization with 2-Player Games **May 2021**
- Extended constrained optimization via 2-player games to a broader class of nonconvex problems (satisfying KL inequality)
- Analyzed stability and convergence of alternating minimax dynamics under nonconvex and nonsmooth constraints

AcaRank **Spring 2018 – Fall 2018**
- Designed and implemented an application ranking academic institutions based on h-index rather than paper quantity

- Use LSTMs and similarity modelling to recommend researchers for review of a paper given abstract

Self-Navigating Robots Use BLE, Circuit Cellar **Dec 2018**
- Designed and constructed a pair of robots that search for a beacon using Bluetooth RSSI

ELI Undergraduate Research Award, Cornell University**,** Ithaca, NY **Oct 2016 – May 2017**
- Supervised by Prof. Eilyan Bitar.
- Designed system for renewable energy in urban environments to dampen fluctuations in energy

WORK EXPERIENCE

Assistantships, University of Illinois at Urbana-Champaign, Urbana, IL **Fall 2021 – current**
- Graduate Research Assistant: research in multitask learning, multi-domain learnings
- Teaching Assistant: CS 446, Machine Learning; develop exams and assignments; manage office hours and logistics
- Teaching Assistant: CS 440, Artificial Intelligence; develop assignments and provide classroom support

Software Development Intern, Microsoft, Remote **May 2021– Aug 2021**
- Built audio ML samples for Windows Machine Learning using ONNX Runtime and DirectML to demo high-performance inference.
- Documented implementation details to provide clear guidance for real-world audio applications.
- Open-source code available at https://github.com/microsoft/Windows-Machine-Learning

Software Development Intern, Microsoft, Redmond, WA **May 2018 – Aug 2018**
- Designed and deployed an automated pipeline using CosmosDB, Azure WebJobs, Azure Active Directory, and other cloud services to collect and compile performance logs from Universal Store microservices.
- Transformed microservice logs to capacity forecast insights, enabling proactive preparation for the holiday shopping surge.
- Strengthened scalability and reliability of Universal Store by ensuring infrastructure could support heavy seasonal traffic loads.

Software Development Intern, Geotab, Oakville, ON **May 2017 – Aug 2017**
- Developed Selenium-based test suites to scrape Geotab web interfaces, generating screenshots to support translation workflows.

Software Development Intern, Geotab, Oakville, ON **May 2016 – Aug 2016**
- Developed scripts to automate creation of virtual machines for dev environments to reduce setup time and ensure consistency.
- Leveraged Google Compute Engine SDK, Big Data (BigQuery) and Google Datalabs to collect and visualize build server status

Teaching Assistantships, Cornell University, Ithaca, NY **Fall 2016 – present**
  - Awards in 2018, 2019, and 2020 for excellence in teaching
  - Average 4.8/5 rating on Teaching Assistantship feedback for Spring 2020
- CS 2112, Honors Object Oriented Programming
- CS 4850: Mathematical Foundations for the Information Age
- CS 4700: Foundations of Artificial Intelligence
- CS 2110: Object Oriented Programming
- CS 4780: Introduction to Machine Learning

SERVICE AND OUTREACH

Conference Reviewer
- International Conference on Machine Learning (ICML) **2023**

Grad Women in CS (GradWCS), UIUC **Aug 2024 – present**

- Coordinated monthly student-faculty luncheons as part of the planning committee, fostering community engagement

Committee Chair: Expand Your Horizons, Cornell University, Ithaca, NY **Aug 2019 – May 2020**

- Facilitated outreach workshops for middle school girls, adapting to remote delivery during COVID-19 to maintain engagement.

Cornell Mars Rover Project Team, Ithaca, NY **Sep 2015 – Aug 2019**

- Programmed microcontroller firmware and designed PCBs for motor controls, enhancing reliability of the Cornell Mars Rover

Women in Computing (WICC) at Cornell **Spring 2017, Fall 2018**

- Mentor undergraduate women interested in computer science

ELI Undergraduate Research Award, Cornell University**,** Ithaca, NY **Oct 2016 – May 2017**

- Awarded for project on renewable energy systems in urban environments for charging electric vehicle fleets

ADDITIONAL SKILLS

Programming: Python, C#, C, C++, Java;

Tools and services: PyTorch, TensorFlow, scikit-learn, SQL, HuggingFace, ONNX Runtime, CUDA